

Hex Inverter

74AC04, 74ACT04

General Description

The AC/ACT04 contains six inverters.

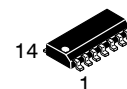
Features

- I_{CC} Reduced by 50% On 74AC Only
- Outputs Source/Sink 24 mA
- ACT04 has TTL-Compatible Inputs
- These are Pb-Free Devices

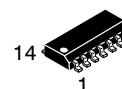
ABSOLUTE MAXIMUM RATINGS

Parameter	Symbol	Value	Unit
Supply Voltage	V_{CC}	-0.5 to +6.5	V
DC Input Diode Current $V_I = -0.5$ V $V_I = V_{CC} + 0.5$ V	I_{IK}	-20 +20	mA
DC Input Voltage	V_I	-0.5 to $V_{CC} + 0.5$	V
DC Output Diode Current $V_O = -0.5$ V $V_O = V_{CC} + 0.5$ V	I_{OK}	-20 +20	mA
DC Output Voltage	V_O	-0.5 to $V_{CC} + 0.5$	V
DC Output Source or Sink Current	I_O	± 50	mA
DC V_{CC} or Ground Current per Output Pin	I_{CC} or I_{GND}	± 50	mA
Storage Temperature Range	T_{STG}	-65 to +150	°C
Junction Temperature	T_J	140	°C

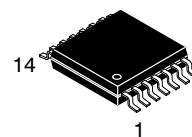
Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.



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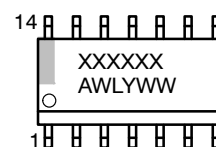


SOIC-14
CASE 751EF



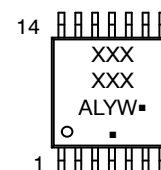
TSSOP-14 WB
CASE 948G

MARKING DIAGRAM



XXXXXX = Specific Device Code
A = Assembly Location
WL = Wafer Lot
Y = Year
WW = Work Week

MARKING DIAGRAM



XXXXXX = Specific Device Code
A = Assembly Location
L = Wafer Lot
Y = Year
W = Work Week
▪ = Pb-Free Package

(Note: Microdot may be in either location)

ORDERING INFORMATION

See detailed ordering and shipping information on page 5 of this data sheet.

74AC04, 74ACT04

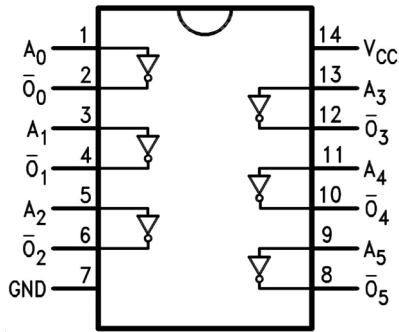


Figure 1. Connection Diagram

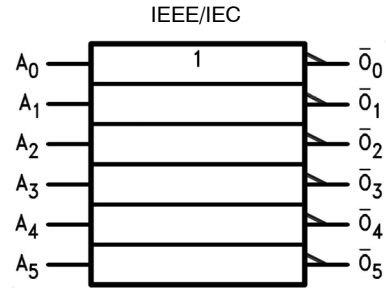


Figure 2. Logic Symbol

PIN DESCRIPTION

Pin	Description
A_n	Inputs
$\bar{O}_n$	Outputs

RECOMMENDED OPERATING CONDITIONS

Symbol	Parameter	Min	Max	Unit
V_{CC}	Supply Voltage AC ACT	2.0 4.5	6.0 5.5	V
V_I	Input Voltage	0	V_{CC}	V
V_O	Output Voltage	0	V_{CC}	V
T_A	Operating Temperature	-40	+85	°C
$\Delta V / \Delta t$	Minimum Input Edge Rate, AC Devices: V_{IN} from 30% to 70% of V_{CC} , V_{CC} at 3.3 V, 4.5 V, 5.5 V	125		mV/ns
$\Delta V / \Delta t$	Minimum Input Edge Rate, ACT Devices: V_{IN} from 0.8 V to 2.0 V, V_{CC} at 4.5 V, 5.5 V	125		mV/ns

Functional operation above the stresses listed in the Recommended Operating Ranges is not implied. Extended exposure to stresses beyond the Recommended Operating Ranges limits may affect device reliability.

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DC ELECTRICAL CHARACTERISTICS FOR AC

Symbol	Parameter	V _{CC} (V)	Conditions	T _A = +25°C		T _A = -40°C to +85°C	Unit
				Typ	Guaranteed Limits		
V _{IH}	Minimum HIGH Level Input Voltage	3.0	V _{OUT} = 0.1 V or V _{CC} - 0.1 V	1.5	2.1	2.1	V
		4.5		2.25	3.15		
		5.5		2.75	3.85		
V _{IL}	Maximum LOW Level Input Voltage	3.0	V _{OUT} = 0.1 V or V _{CC} - 0.1 V	1.5	0.9	0.9	V
		4.5		2.25	1.35		
		5.5		2.75	1.65		
V _{OH}	Minimum HIGH Level Output Voltage	3.0	I _{OUT} = -50 μA	2.99	2.9	2.9	V
		4.5		4.49	4.4	4.4	
		5.5		5.49	5.4	5.4	
		3.0	V _{IN} = V _{IL} or V _{IH} , I _{OH} = -12 mA	-	2.56	2.46	
		4.5	I _{OH} = -24 mA	-	3.86	3.76	
		5.5	I _{OH} = -24 mA (Note 1)	-	4.86	4.76	
V _{OL}	Maximum LOW Level Output Voltage	3.0	I _{OUT} = 50 μA	0.002	0.1	0.1	V
		4.5		0.001	0.1	0.1	
		5.5		0.001	0.1	0.1	
		3.0	V _{IN} = V _{IL} or V _{IH} , I _{OL} = 12 mA	-	0.36	0.44	
		4.5	I _{OL} = 24 mA	-	0.36	0.44	
		5.5	I _{OL} = 24 mA (Note 1)	-	0.36	0.44	
I _{IN} (Note 3)	Maximum Input Leakage Current	5.5	V _I = V _{CC} , GND	-	±0.1	±1.0	μA
I _{OLD}	Minimum Dynamic Output Current (Note 2)	5.5	V _{OLD} = 1.65 V Max.	-	-	75	mA
I _{OHD}		5.5	V _{OHD} = 3.85 V Min.	-	-	-75	mA
I _{CC} (Note 3)	Maximum Quiescent Supply Current	5.5	V _{IN} = V _{CC} or GND	-	2.0	20.0	μA

1. All outputs loaded; thresholds on input associated with output under test.

2. Maximum test duration 2.0 ms, one output loaded at a time.

3. I_{IN} and I_{CC} at 3.0 V are guaranteed to be less than or equal to the respective limit at 5.5 V V_{CC}.

74AC04, 74ACT04

DC ELECTRICAL CHARACTERISTICS FOR ACT

Symbol	Parameter	V _{CC} (V)	Conditions	T _A = +25°C		T _A = -40°C to +85°C	Unit
				Typ	Guaranteed Limits		
V _{IH}	Minimum HIGH Level Input Voltage	4.5	V _{OUT} = 0.1 V or V _{CC} - 0.1 V	1.5	2.0	2.0	V
		5.5		1.5	2.0	2.0	
V _{IL}	Maximum LOW Level Input Voltage	4.5	V _{OUT} = 0.1 V or V _{CC} - 0.1 V	1.5	0.8	0.8	V
		5.5		1.5	0.8	0.8	
V _{OH}	Minimum HIGH Level Output Voltage	4.5	I _{OUT} = -50 μA	4.49	4.4	4.4	V
		5.5		5.49	5.4	5.4	
		4.5	V _{IN} = V _{IL} or V _{IH} I _{OH} = -24 mA	–	3.86	3.76	
		5.5	I _{OH} = -24 mA (Note 4)	–	4.86	4.76	
V _{OL}	Maximum LOW Level Output Voltage	4.5	I _{OUT} = 50 μA	0.001	0.1	0.1	V
		5.5		0.001	0.1	0.1	
		4.5	V _{IN} = V _{IL} or V _{IH} I _{OL} = 24 mA	–	0.36	0.44	
		5.5	I _{OL} = 24 mA (Note 4)	–	0.36	0.44	
I _{IN}	Maximum Input Leakage Current	5.5	V _I = V _{CC} , GND	–	±0.1	±1.0	μA
I _{CCT}	Maximum I _{CC} /Input	5.5	V _I = V _{CC} - 2.1 V	0.6	–	1.5	mA
I _{OLD}	Minimum Dynamic Output Current (Note 5)	5.5	V _{OLD} = 1.65 V Max.	–	–	75	mA
I _{OHD}		5.5	V _{OHD} = 3.85 V Min.	–	–	–75	mA
I _{CC}	Maximum Quiescent Supply Current	5.5	V _{IN} = V _{CC} or GND	–	4.0	40.0	μA

4. All outputs loaded; thresholds on input associated with output under test.

5. Maximum test duration 2.0 ms, one output loaded at a time.

74AC04, 74ACT04

AC ELECTRICAL CHARACTERISTICS FOR AC

Symbol	Parameter	V _{CC} (V) (Note 6)	T _A = +25°C, C _L = 50 pF			T _A = -40°C to +85°C, C _L = 50 pF		Unit
			Min	Typ	Max	Min	Max	
t _{PLH}	Propagation Delay	3.3	1.5	4.5	9.0	1.0	10.0	ns
		5.0	1.5	4.0	7.0	1.0	7.5	
t _{PHL}	Propagation Delay	3.3	1.5	4.5	8.5	1.0	9.5	ns
		5.0	1.5	3.5	6.5	1.0	7.0	

6. Voltage range 3.3 is 3.3 V + 0.3 V. Voltage range 5.0 is 5.0 V + 0.5 V.

AC ELECTRICAL CHARACTERISTICS FOR ACT

Symbol	Parameter	V _{CC} (V) (Note 7)	T _A = +25°C, C _L = 50 pF			T _A = -40°C to +85°C, C _L = 50 pF		Unit
			Min	Typ	Max	Min	Max	
t _{PLH}	Propagation Delay	5.0	1.0	6.0	8.5	1.0	9.0	ns
t _{PLH}	Propagation Delay	5.0	1.0	5.5	8.0	1.0	8.5	ns

7. Voltage range 5.0 is 5.0 V + 0.5 V.

CAPACITANCE

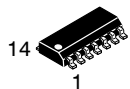
Symbol	Parameter	Conditions	Typ	Unit
C _{IN}	Input Capacitance	V _{CC} = OPEN	4.5	pF
V _{CC}	Power Dissipation Capacitance	V _{CC} = 5.0 V	30.0	pF

ORDERING INFORMATION

Order Number	Marking	Package	Shipping [†]
74AC04SC	AC04G	SOIC-14	55 Units / Rail
74AC04SCX	AC04G	SOIC-14	2500 / Tape & Reel
74ACT04MTC	ACT 04	TSSOP-14	96 Units / Rail
74AC04MTCX	AC 04	TSSOP-14	2500 / Tape & Reel
74ACT04SC	ACT04G	SOIC-14	55 Units / Rail
74ACT04SCX	ACT04G	SOIC-14	2500 / Tape & Reel
74ACT04MTCX	ACT 04	TSSOP-14	2500 / Tape & Reel

[†]For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

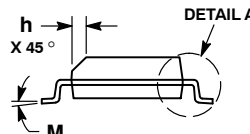
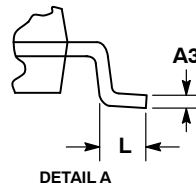
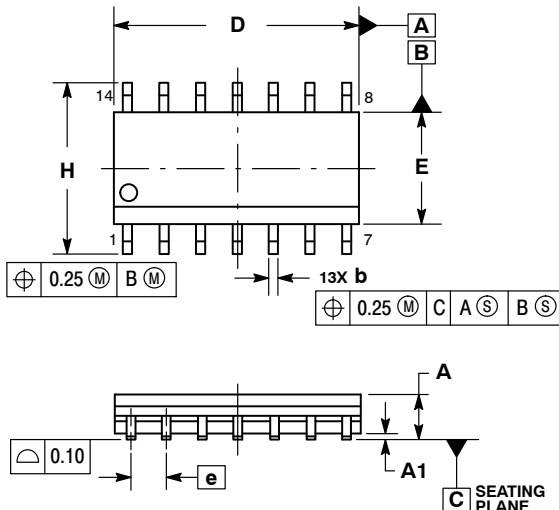
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SCALE 1:1

SOIC-14 NB
CASE 751A-03
ISSUE L

DATE 03 FEB 2016

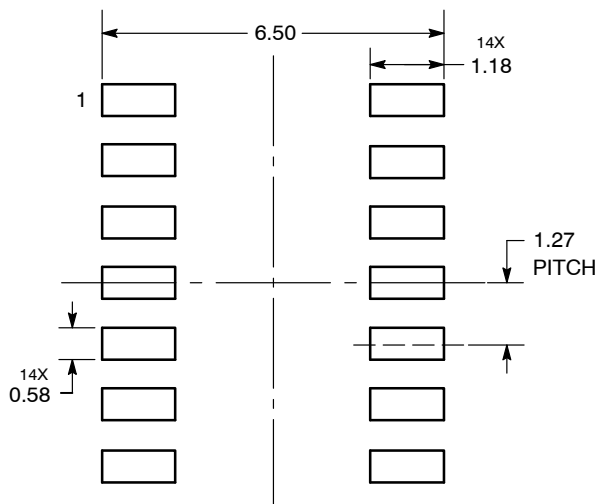


NOTES:

1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 1994.
2. CONTROLLING DIMENSION: MILLIMETERS.
3. DIMENSION b DOES NOT INCLUDE DAMBAR PROTRUSION. ALLOWABLE PROTRUSION SHALL BE 0.13 TOTAL IN EXCESS OF AT MAXIMUM MATERIAL CONDITION.
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD PROTRUSIONS.
5. MAXIMUM MOLD PROTRUSION 0.15 PER SIDE.

DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	1.35	1.75	0.054	0.068
A1	0.10	0.25	0.004	0.010
A3	0.19	0.25	0.008	0.010
b	0.35	0.49	0.014	0.019
D	8.55	8.75	0.337	0.344
E	3.80	4.00	0.150	0.157
e	1.27 BSC		0.050 BSC	
H	5.80	6.20	0.228	0.244
h	0.25	0.50	0.010	0.019
L	0.40	1.25	0.016	0.049
M	0°	7°	0°	7°

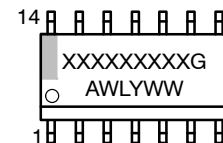
SOLDERING FOOTPRINT*



DIMENSIONS: MILLIMETERS

*For additional information on our Pb-Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

GENERIC
MARKING DIAGRAM*



XXXXXX = Specific Device Code
A = Assembly Location
WL = Wafer Lot
Y = Year
WW = Work Week
G = Pb-Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb-Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.

STYLES ON PAGE 2

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CASE 751A-03
ISSUE L

DATE 03 FEB 2016

STYLE 1:
PIN 1. COMMON CATHODE
2. ANODE/CATHODE
3. ANODE/CATHODE
4. NO CONNECTION
5. ANODE/CATHODE
6. NO CONNECTION
7. ANODE/CATHODE
8. ANODE/CATHODE
9. ANODE/CATHODE
10. NO CONNECTION
11. ANODE/CATHODE
12. ANODE/CATHODE
13. NO CONNECTION
14. COMMON ANODE

STYLE 2:
CANCELLED

STYLE 3:
PIN 1. NO CONNECTION
2. ANODE
3. ANODE
4. NO CONNECTION
5. ANODE
6. NO CONNECTION
7. ANODE
8. ANODE
9. ANODE
10. NO CONNECTION
11. ANODE
12. ANODE
13. NO CONNECTION
14. COMMON CATHODE

STYLE 4:
PIN 1. NO CONNECTION
2. CATHODE
3. CATHODE
4. NO CONNECTION
5. CATHODE
6. NO CONNECTION
7. CATHODE
8. CATHODE
9. CATHODE
10. NO CONNECTION
11. CATHODE
12. CATHODE
13. NO CONNECTION
14. COMMON ANODE

STYLE 5:
PIN 1. COMMON CATHODE
2. ANODE/CATHODE
3. ANODE/CATHODE
4. ANODE/CATHODE
5. ANODE/CATHODE
6. NO CONNECTION
7. COMMON ANODE
8. COMMON CATHODE
9. ANODE/CATHODE
10. ANODE/CATHODE
11. ANODE/CATHODE
12. ANODE/CATHODE
13. NO CONNECTION
14. COMMON ANODE

STYLE 6:
PIN 1. CATHODE
2. CATHODE
3. CATHODE
4. CATHODE
5. CATHODE
6. CATHODE
7. CATHODE
8. ANODE
9. ANODE
10. ANODE
11. ANODE
12. ANODE
13. ANODE
14. ANODE

STYLE 7:
PIN 1. ANODE/CATHODE
2. COMMON ANODE
3. COMMON CATHODE
4. ANODE/CATHODE
5. ANODE/CATHODE
6. ANODE/CATHODE
7. ANODE/CATHODE
8. ANODE/CATHODE
9. ANODE/CATHODE
10. ANODE/CATHODE
11. COMMON CATHODE
12. COMMON ANODE
13. ANODE/CATHODE
14. ANODE/CATHODE

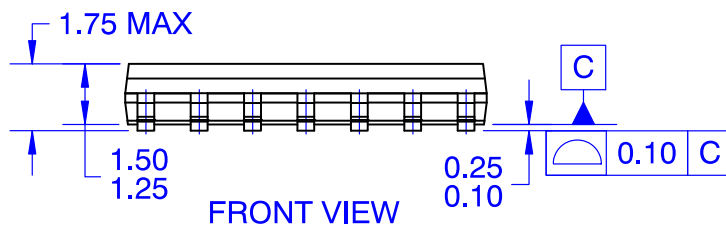
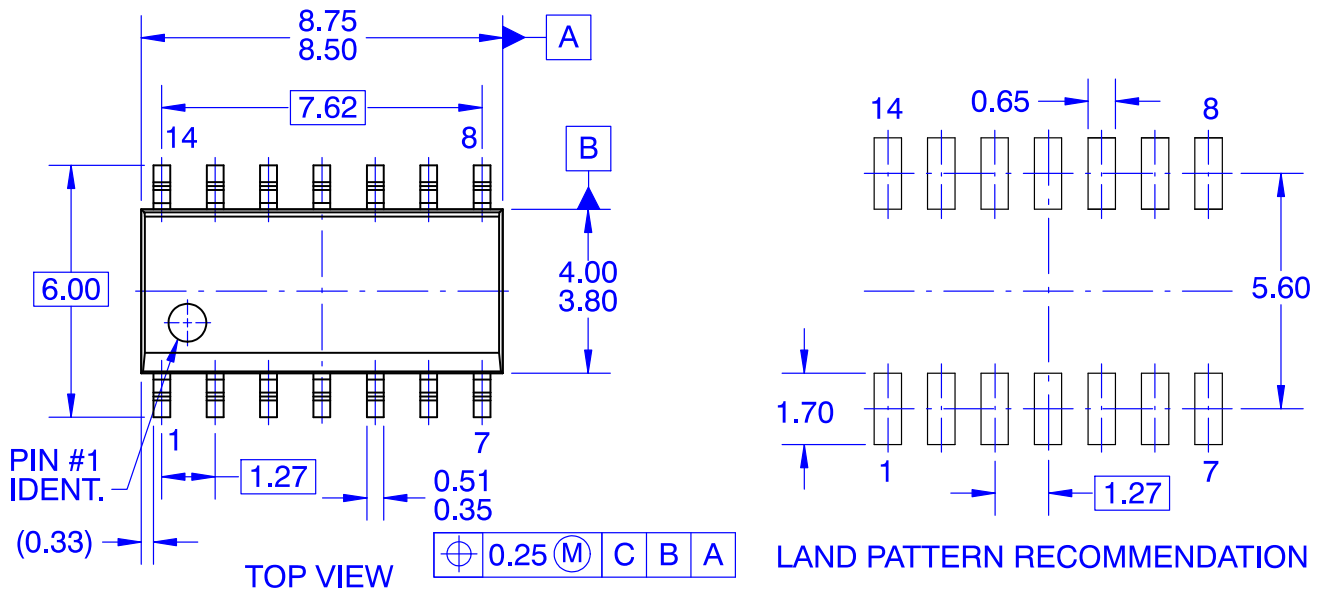
STYLE 8:
PIN 1. COMMON CATHODE
2. ANODE/CATHODE
3. ANODE/CATHODE
4. NO CONNECTION
5. ANODE/CATHODE
6. ANODE/CATHODE
7. COMMON ANODE
8. COMMON ANODE
9. ANODE/CATHODE
10. ANODE/CATHODE
11. NO CONNECTION
12. ANODE/CATHODE
13. ANODE/CATHODE
14. COMMON CATHODE

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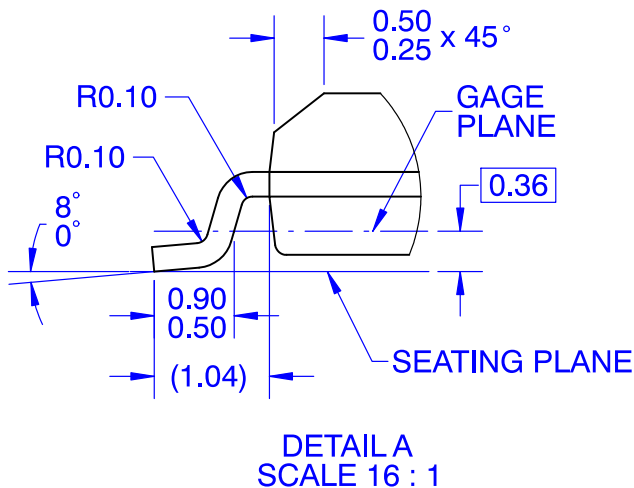
SOIC14
CASE 751EF
ISSUE O

DATE 30 SEP 2016



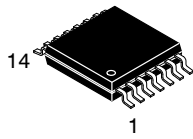
NOTES:

- A. CONFORMS TO JEDEC MS-012, VARIATION AB, ISSUE C
- B. ALL DIMENSIONS ARE IN MILLIMETERS
- C. DIMENSIONS DO NOT INCLUDE MOLD FLASH OR BURRS
- D. LAND PATTERN STANDARD: SOIC127P600X145-14M
- E. CONFORMS TO ASME Y14.5M, 2009

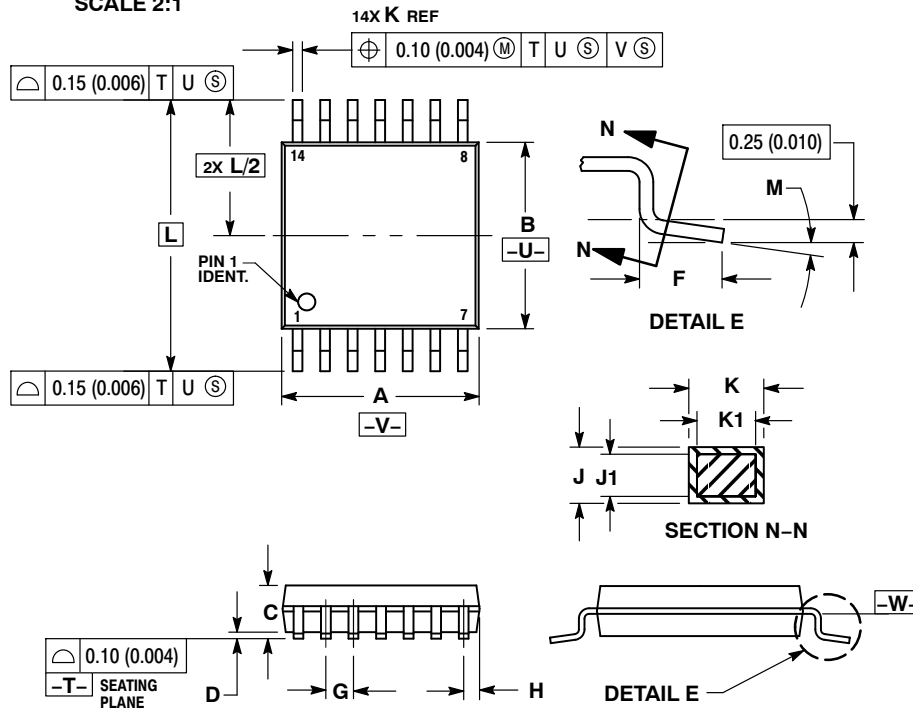


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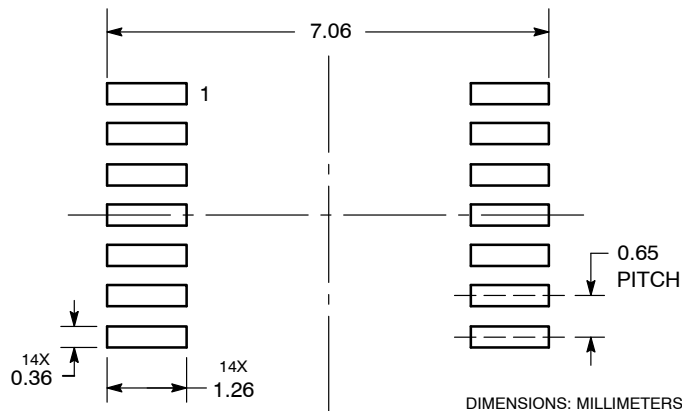

TSSOP-14 WB
CASE 948G
ISSUE C

DATE 17 FEB 2016

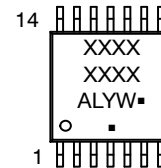

NOTES:

1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: MILLIMETER.
3. DIMENSION A DOES NOT INCLUDE MOLD FLASH, PROTRUSIONS OR GATE BURRS. MOLD FLASH OR GATE BURRS SHALL NOT EXCEED 0.15 (0.006) PER SIDE.
4. DIMENSION B DOES NOT INCLUDE INTERLEAD FLASH OR PROTRUSION. INTERLEAD FLASH OR PROTRUSION SHALL NOT EXCEED 0.25 (0.010) PER SIDE.
5. DIMENSION K DOES NOT INCLUDE DAMBAR PROTRUSION. ALLOWABLE DAMBAR PROTRUSION SHALL BE 0.08 (0.003) TOTAL IN EXCESS OF THE K DIMENSION AT MAXIMUM MATERIAL CONDITION.
6. TERMINAL NUMBERS ARE SHOWN FOR REFERENCE ONLY.
7. DIMENSION A AND B ARE TO BE DETERMINED AT DATUM PLANE -W-.

DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	4.90	5.10	0.193	0.200
B	4.30	4.50	0.169	0.177
C	---	1.20	---	0.047
D	0.05	0.15	0.002	0.006
F	0.50	0.75	0.020	0.030
G	0.65 BSC		0.026 BSC	
H	0.50	0.60	0.020	0.024
J	0.09	0.20	0.004	0.008
J1	0.09	0.16	0.004	0.006
K	0.19	0.30	0.007	0.012
K1	0.19	0.25	0.007	0.010
L	6.40 BSC		0.252 BSC	
M	0°	8°	0°	8°

**RECOMMENDED
SOLDERING FOOTPRINT***


*For additional information on our Pb-Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

**GENERIC
MARKING DIAGRAM***


A = Assembly Location
L = Wafer Lot
Y = Year
W = Work Week
■ = Pb-Free Package

(Note: Microdot may be in either location)

*This information is generic. Please refer to device data sheet for actual part marking. Pb-Free indicator, "G" or microdot "■", may or may not be present. Some products may not follow the Generic Marking.

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